Theorem

$$QV\left(W_t, [a,b]\right) \xrightarrow[L^2]{m.s} (b-a)$$

Remember

Convergence in mean square

The sequence ***converges in mean square*** *to* X, *shown by* $X_n \xrightarrow{m.s.} X$, *if*

$$\lim_{n\to\infty} E\left((X_n - X)^2\right) = 0.$$

Here $X_n \equiv Q_2^{(n)}$ and $X \equiv b-a$

Proof We need to show that

$$\lim_{\|\Pi_n\|\to 0} E\left[\left(Q_2^{(n)}(W_t) - (b-a)\right)^2\right] = 0$$

$$\lim_{\|\Pi_n\|\to 0} E\left[\underbrace{\left(\sum_{i=0}^{n-1} |W_{t_{i+1}} - W_{t_i}|^2 - (b-a)\right)}_{Y_n}{}^2\right] = 0$$

Note that

$$Y_n = \sum_{i=0}^{n-1} \underbrace{\left(|W_{t_{i+1}} - W_{t_i}|^2 - (t_{i+1} - t_i)\right)}_{Z_i \text{ (depends on } \Pi_n)}$$

with $E(Y_n) = 0$ HW

Also, note that

$$Y_n^2 = \sum_{i=0}^{n-1} Z_i^2 + 2\sum_{i<j} Z_i Z_j$$

and

(HW)

$$E(Y_n^2) = \sum_{i=0}^{n-1} E(Z_i^2) + 2\sum_{i<j} \cancel{E(Z_i Z_j)} \to 0$$

$$= \sum \cancel{E\left[\left(W_{t_{i+1}} - W_{t_i}\right)^4\right]} \to 3(t_{i+1}-t_i)^2 \quad \text{HW}$$

$$- 2(t_{i+1}-t_i)\cancel{E\left[\left(W_{t_{i+1}} - W_{t_i}\right)^2\right]} \to (t_{i+1}-t_i) \quad \text{HW}$$

$$+ (t_{i+1}-t_i)^2$$

$$= \sum 3(t_{i+1}-t_i)^2 - 2(t_{i+1}-t_i)^2 + (t_{i+1}-t_i)^2$$

$$= \sum_{i=0}^{n-1} 2(t_{i+1}-t_i)^2 \leq 2\|\Pi_n\| \sum_{i=0}^{n-1} (t_{i+1}-t_i)$$

$$= 2(b-a)\|\Pi_n\| \xrightarrow{m.s} 0 \quad \text{as } n \to \infty$$

or $\|\Pi_n\| \to 0$

Symbolically, we write

$$(dW_t)^2 = dt$$

Theorem

$$Q_2^{(n)}(W_t) \xrightarrow{a.s} (b-a)$$

Use Chebychev's inequality and Borel-Cantelli Lemma.

Proof in last page

First Variation

$$Q_1^{(n)}(W) \xrightarrow{a.s} \infty$$

We proved that

$$Q_2^{(n)} \xrightarrow{a.s} (b-a)$$

But

$$\lim_{\|\pi\| \to 0} \sum_j \left(W(t_{j+1}) - W(t_j)\right)^2$$

$$\leq \lim_{\|\pi_n\| \to 0} \sum_j \left|W(t_{j+1}) - W(t_j)\right| \, |\Delta W|_{max}$$

$$\leq \lim_{\|\pi_n\| \to 0} |\Delta W|_{max} \underbrace{\sum_j \left|W(t_{j+1}) - W(t_j)\right|}_{Q_1^{(n)}(W_t)}$$

$|\Delta W|_{max} \to 0$ W_t is continuous

$Q_1^{(n)}(W_t) \to \infty$ must go to inf

otherwise Q_2 cannot be $(b-a)$

HW

① Prove that

$$QV\left(W_t, [a,b]\right) \xrightarrow[L^2]{m.s} (b-a)$$

by showing that

$$E\left(Q_2^{(n)}\right) \to b-a \quad \text{and} \quad Var\left(Q_2^{(n)}\right) \to 0$$

as $\|\Pi_n\| \to 0$. Explain your work

② Show that

a) $$\lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} \left(t_{j+1} - t_j\right)^2 = 0$$

or $$(dt)^2 = 0$$

b) $$\lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} \left(W_{t_{j+1}} - W_{t_j}\right)\left(t_{j+1} - t_j\right) = 0$$

or $$dW_t\, dt = 0$$

Theorem

$$Q_2^{(n)}(W_t) \xrightarrow{a.s.} (b-a)$$

Remember

Almost sure convergence

The sequence converges to X ***almost surely****, shown by* $X_n \xrightarrow{a.s.} X$*, if*

$$\mathbb{P}\Big(\underbrace{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)}_{A.}\Big) = 1.$$

In some cases it is very difficult to define A.

The Borel-Cantelli lemma. is very powerful alternative:

Let $A_n = \{ w : |X_n(w) - X(w)| > \varepsilon \}$

If $\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} P(|X_n - X| > \varepsilon) < \infty$

then $X_n \xrightarrow{a.s.} X$

Verify that $E(Y_n) = 0$ and

$$Var(Y_n) = E(Y_n^2) = 2\sum_i (t_{i+1} - t_i)^2$$

Use Chebychev's inequality

Chebyshev's inequality

If X is a random variable with mean μ and variance σ^2, then for any $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

$$P(|Y_n| > \varepsilon) = P\left(|Y_n| > \frac{\varepsilon}{\sqrt{Var}}\sqrt{Var}\right)$$

$$\leq \frac{Var}{\varepsilon^2} = \frac{2}{\varepsilon^2}\sum_i (t_{i+1} - t_i)^2$$

$$\leq \frac{2}{\varepsilon^2}\sum_{i=0}^{n-1}(t_{i+1} - t_i)\|\Pi_n\| = \frac{2\|\Pi_n\|}{\varepsilon^2}(b-a) \to 0$$

as $n \to \infty$,

So,

$$\sum_{n=1}^{\infty} P(|Y_n| > \varepsilon) = \frac{2(b-a)}{\varepsilon^2}\underbrace{\sum_{n=1}^{\infty}\|\Pi_n\|}_{<\infty} < \infty$$

Therefore $Y_n \xrightarrow{a.s.} 0$ or $Q_2^{(n)} \xrightarrow{a.s.} (b-a)$

or

$$QV(W_t) \xrightarrow{a.s} (b-a)$$